

Differential Flatness and Path Generation

A Comprehensive Mathematical Tutorial for PMSM Control

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Abstract

This tutorial provides a self-contained and rigorous mathematical introduction to differential flatness, with a strong emphasis on its application to permanent magnet synchronous motors (PMSM). We begin with the formal definition of differential flatness and the underlying differential-geometric concepts, including Lie derivatives and the role of the vector relative degree. We then prove that the PMSM is flat with the rotor mechanical angle as the flat output. The flatness mapping is derived in full detail, showing how the flat output and its derivatives determine the state and input variables. Next, we address path generation: the construction of smooth jerk-limited trajectories that serve as the reference for the flat output. The stopping acceleration formula is proved, and the discrete-time integration scheme is derived from constant-jerk kinematics. The relation between flatness and feedback linearization is discussed, and the practical control architecture—combining feedforward and PI feedback—is presented as a natural consequence of the theory. The tutorial is self-contained and includes all necessary proofs.

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1 Introduction

Differential flatness is a structural property of a class of nonlinear dynamical systems that greatly simplifies control design, especially for trajectory tracking and feedforward control. The central idea is that a specially chosen output, called a *flat output*, and its derivatives can be used to parameterize all system states and inputs without solving differential equations. This is in contrast to classical control methods, where one must often solve differential equations to determine the required inputs for a desired output trajectory.

For a permanent magnet synchronous motor (PMSM), the flat output is the mechanical rotor angle. Once a smooth desired rotor angle trajectory is specified, the required currents and voltages can be computed analytically using the flatness equations. This is the approach taken in the EmbedSim DFC controller.

This tutorial is structured as follows:

- Section 2 gives the formal definition of differential flatness and introduces Lie derivatives.
- Section 3 proves that the PMSM is flat with the mechanical angle as the flat output.
- Section 4 derives the full flatness mapping from the flat output to the control inputs.
- Section 5 explains path generation, including the jerk-limited trajectory algorithm and the stopping acceleration formula.
- Section 5.5 presents the discrete-time integration scheme and its derivation.
- Section 6 discusses the relationship between flatness and feedback linearization.
- Section 7 describes the practical control architecture that combines feedforward and feedback.

Throughout, we emphasize the mathematical foundations and provide proofs.

2 Differential Flatness: Definition and Lie Derivatives

Consider a nonlinear control system in state-space form:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x} \in \mathbb{R}^n, \quad \mathbf{u} \in \mathbb{R}^m. \quad (1)$$

For control-affine systems, which are common in physical applications, the dynamics take the form:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \sum_{i=1}^m \mathbf{g}_i(\mathbf{x})u_i, \quad (2)$$

where $\mathbf{f}, \mathbf{g}_i$ are smooth vector fields.

2.1 Definition of Differential Flatness

Definition 2.1 (Differentially Flat System) *The system (??) is said to be differentially flat if there exists an output*

$$\mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}, \dot{\mathbf{u}}, \dots, \mathbf{u}^{(\alpha)}), \quad (3)$$

with components $y_1, \dots, y_m$ (same dimension as the input), such that:

1. *The state $\mathbf{x}$ and input $\mathbf{u}$ can be expressed as functions of $\mathbf{y}$ and a finite number of its derivatives:*

$$\mathbf{x} = \Phi_x(\mathbf{y}, \dot{\mathbf{y}}, \ddot{\mathbf{y}}, \dots, \mathbf{y}^{(\beta)}), \quad (4)$$

$$\mathbf{u} = \Phi_u(\mathbf{y}, \dot{\mathbf{y}}, \ddot{\mathbf{y}}, \dots, \mathbf{y}^{(\beta+1)}). \quad (5)$$

2. The components of $\mathbf{y}$ are differentially independent: no nontrivial differential relation exists among them, i.e., the map from $\mathbf{y}$ and its derivatives to $\mathbf{x}$ and $\mathbf{u}$ is a local diffeomorphism.

The property of differential flatness is important because it allows the direct design of a reference trajectory in the flat-output space, which automatically yields the required state and input trajectories.

2.2 Lie Derivatives

To verify flatness and to find a flat output, we often use the concept of *Lie derivatives*. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth scalar function, and let $\mathbf{f}$ be a smooth vector field.

Definition 2.2 (Lie Derivative) The Lie derivative of h along $\mathbf{f}$ is defined as:

$$L_{\mathbf{f}}h(\mathbf{x}) = \frac{\partial h}{\partial \mathbf{x}} \mathbf{f}(\mathbf{x}) = \nabla h \cdot \mathbf{f}. \quad (6)$$

Higher-order Lie derivatives are defined recursively:

$$L_{\mathbf{f}}^k h = L_{\mathbf{f}} \left(L_{\mathbf{f}}^{k-1} h \right), \quad L_{\mathbf{f}}^0 h = h. \quad (7)$$

For a vector field $\mathbf{g}$, the Lie derivative of h along $\mathbf{g}$ is similarly:

$$L_{\mathbf{g}}h = \nabla h \cdot \mathbf{g}. \quad (8)$$

For a control-affine system, the derivative of the output $y = h(\mathbf{x})$ along the system dynamics is:

$$\dot{y} = L_{\mathbf{f}}h(\mathbf{x}) + \sum_{i=1}^m L_{\mathbf{g}_i}h(\mathbf{x})u_i. \quad (9)$$

Differentiating repeatedly, we get:

$$y^{(r)} = L_{\mathbf{f}}^r h + \sum_{i=1}^m L_{\mathbf{g}_i} L_{\mathbf{f}}^{r-1} h u_i + \text{terms involving derivatives of } \mathbf{u}. \quad (10)$$

When the input appears for the first time at some derivative r , the coefficient matrix $L_{\mathbf{g}_i} L_{\mathbf{f}}^{r-1} h$ must be invertible. This is the *vector relative degree* condition.

If the system is flat, there exists an output such that the relative degree is equal to the state dimension (after including derivatives), and the mapping from the output derivatives to the state and input is invertible.

2.3 Illustrative Example: Mass-Spring-Damper

Consider the system:

$$\ddot{x} + b\dot{x} + kx = u. \quad (11)$$

In state-space form with $x_1 = x, x_2 = \dot{x}$, we have $\dot{x}_1 = x_2, \dot{x}_2 = -kx_1 - bx_2 + u$. Choose $y = x_1$. Then:

$$\dot{y} = x_2, \quad \ddot{y} = -ky - b\dot{y} + u. \quad (12)$$

Solving for u : $u = \ddot{y} + b\dot{y} + ky$. The state is $(y, \dot{y})$, and the input is a function of $(y, \dot{y}, \ddot{y})$. Thus the system is flat with flat output $y = x$.

3 Proof That the PMSM Is Flat

We consider a surface-mount PMSM (or a general PMSM operated with $i_d = 0$). The dynamics in the dq reference frame are:

$$\dot{\theta}_m = \omega_m, \quad (13)$$

$$J\dot{\omega}_m = K_t i_q - B\omega_m - T_L, \quad (14)$$

$$L_d \dot{i}_d = v_d - R_s i_d + \omega_e L_q i_q, \quad (15)$$

$$L_q \dot{i}_q = v_q - R_s i_q - \omega_e (L_d i_d + \lambda_{pm}), \quad (16)$$

where $\omega_e = p\omega_m$, $K_t = \frac{3}{2}p\lambda_{pm}$, and p is the number of pole pairs.

We propose the flat output:

$$y = \theta_m. \quad (17)$$

We will show that all states and inputs can be expressed as functions of y and its derivatives up to third order.

3.1 State Reconstruction

From (13), we have:

$$\omega_m = \dot{y}. \quad (18)$$

From (14), solving for i_q :

$$i_q = \frac{J\ddot{y} + B\dot{y} + T_L}{K_t}. \quad (19)$$

Assuming $i_d = 0$ (which is the desired operation for surface PMSM), we have:

$$i_d = 0. \quad (20)$$

Thus the state $(\theta_m, \omega_m, i_d, i_q)$ is completely determined by $y, \dot{y}, \ddot{y}$.

3.2 Input Reconstruction

From (16), we need $\dot{i}_q$. Differentiating (19):

$$\dot{i}_q = \frac{J\ddot{\dot{y}} + B\ddot{y}}{K_t}. \quad (21)$$

Now from the q-axis voltage equation (16) with $i_d = 0$:

$$v_q = R_s i_q + L_q \dot{i}_q + \omega_e \lambda_{pm}. \quad (22)$$

Substitute $\omega_e = p\dot{y}$, i_q from (19), and $\dot{i}_q$ from (21):

$$v_q = R_s \frac{J\ddot{y} + B\dot{y} + T_L}{K_t} + L_q \frac{J\ddot{\dot{y}} + B\ddot{y}}{K_t} + p\dot{y}\lambda_{pm}. \quad (23)$$

From the d-axis voltage equation (15) with $i_d = 0$ and $\dot{i}_d = 0$, we get:

$$v_d = -\omega_e L_q i_q = -p\dot{y} L_q \frac{J\ddot{y} + B\dot{y} + T_L}{K_t}. \quad (24)$$

Thus the inputs (v_d, v_q) are also determined by y and its derivatives up to third order.

3.3 Formal Statement

Theorem 3.1 (Flatness of the PMSM) *The PMSM model (13)–(16) is differentially flat with flat output $y = \theta_m$. The state and input are given by (19), (21), (23), (24), with $i_d = 0$.*

We have explicitly constructed the maps Φ_x and Φ_u :

$$\Phi_x(y, \dot{y}, \ddot{y}) = \left(y, \dot{y}, 0, \frac{J\ddot{y} + B\dot{y} + T_L}{K_t} \right), \quad (25)$$

$$\Phi_u(y, \dot{y}, \ddot{y}, \ddot{\ddot{y}}) = \left(\begin{array}{c} -p\dot{y}L_q \frac{J\ddot{y} + B\dot{y} + T_L}{K_t} \\ R_s \frac{J\ddot{y} + B\dot{y} + T_L}{K_t} + L_q \frac{J\ddot{\ddot{y}} + B\ddot{y}}{K_t} + p\dot{y}\lambda_{pm} \end{array} \right). \quad (26)$$

These mappings are smooth and invertible (the Jacobian has full rank) for $K_t \neq 0$, which holds since $\lambda_{pm} > 0$ and $p > 0$. Hence the system is flat.

4 The Flatness Mapping in Detail

In practice, we do not plan the angle y directly; we plan the desired velocity ω^* and then integrate to get θ^* . The flatness equations are most conveniently expressed in terms of ω^* , its derivative (acceleration) $\dot{\omega}^*$, and its second derivative (jerk) $\ddot{\omega}^*$.

Let:

$$\omega^* = \dot{y}, \quad \dot{\omega}^* = \ddot{y}, \quad \ddot{\omega}^* = \ddot{\ddot{y}}. \quad (27)$$

Then the state and input reconstruction becomes:

4.1 Mechanical Torque

The required torque to follow ω^* is obtained from (14):

$$T_{ff} = J\dot{\omega}^* + B\omega^* + T_L. \quad (28)$$

This is the feedforward torque.

4.2 Current Reference

From (19):

$$i_q^* = \frac{J\dot{\omega}^* + B\omega^* + T_L}{K_t} = \frac{T_{ff}}{K_t}. \quad (29)$$

4.3 Current Derivative Feedforward

From (21):

$$\dot{i}_q^* = \frac{J\ddot{\omega}^* + B\dot{\omega}^*}{K_t}. \quad (30)$$

4.4 Voltage Feedforward

From (24) and (23):

$$v_d^* = -p\omega^*L_q\dot{i}_q^*, \quad (31)$$

$$v_q^* = R_s\dot{i}_q^* + L_q\ddot{i}_q^* + p\omega^*\lambda_{pm}. \quad (32)$$

These are the feedforward voltages.

4.5 Feedback Corrections

In the practical controller, speed and current PI loops add corrections to handle modelling errors and disturbances:

$$T^* = T_{ff} + \Delta T, \quad \Delta T = K_{p,\omega} e_\omega + K_{i,\omega} \int e_\omega, \quad (33)$$

$$i_q^* = \frac{T^*}{K_t}, \quad (34)$$

$$v_d = v_d^* + \Delta v_d, \quad \Delta v_d = K_{p,d} e_d + K_{i,d} \int e_d, \quad (35)$$

$$v_q = v_q^* + \Delta v_q, \quad \Delta v_q = K_{p,q} e_q + K_{i,q} \int e_q. \quad (36)$$

5 Path Generation: Constructing the Trajectory

The flatness mapping requires a smooth reference trajectory $\omega^*(t)$, $\dot{\omega}^*(t)$, $\ddot{\omega}^*(t)$. These are generated by an online trajectory planner that respects physical limits on acceleration and jerk.

5.1 Problem Statement

Given:

- Current measured velocity ω ,
- Target velocity ω_{target} ,
- Maximum acceleration $a_{\text{max}} > 0$,
- Maximum jerk $j_{\text{max}} > 0$,
- Sample time $T_s > 0$,
- Previous acceleration a_{prev} ,

compute the next desired velocity ω^* , acceleration $\dot{\omega}^*$, and jerk $\ddot{\omega}^*$ such that:

1. $|\ddot{\omega}^*| \leq j_{\text{max}}$,
2. $|\dot{\omega}^*| \leq a_{\text{max}}$,
3. The velocity approaches ω_{target} smoothly.

5.2 Derivation of the Stopping Acceleration

Let $e = \omega_{\text{target}} - \omega$ be the velocity error. Suppose we decelerate at maximum jerk magnitude j_{max} from acceleration a down to zero, then continue with zero acceleration until the velocity error is eliminated. The velocity change during the triangular deceleration phase is:

$$\Delta\omega_{\text{decel}} = \frac{a^2}{2j_{\text{max}}}. \quad (37)$$

Setting $\Delta\omega_{\text{decel}} = |e|$ gives the stopping acceleration:

$$a_{\text{stop}} = \sqrt{2j_{\text{max}}|e|}. \quad (38)$$

This is the maximum acceleration that can be used to remove the velocity error without exceeding the jerk limit.

5.3 Desired Acceleration

The desired acceleration is then:

$$a_{\text{des}} = \text{sgn}(e) \min(a_{\text{max}}, a_{\text{stop}}). \quad (39)$$

This ensures that we accelerate in the correct direction, with magnitude limited by both the hardware limit a_{max} and the stopping acceleration.

5.4 Jerk Calculation

Given the previous acceleration a_{prev} , the jerk required to reach a_{des} in one sample is:

$$j_{\text{req}} = \frac{a_{\text{des}} - a_{\text{prev}}}{T_s}. \quad (40)$$

Saturate:

$$j = \text{sat}(j_{\text{req}}, -j_{\text{max}}, j_{\text{max}}). \quad (41)$$

5.5 Discrete-Time Integration

Assuming constant jerk j during the sample period, we integrate:

$$a_{\text{new}} = a_{\text{prev}} + jT_s, \quad (42)$$

$$\omega_{\text{new}} = \omega + a_{\text{prev}}T_s + \frac{1}{2}jT_s^2, \quad (43)$$

$$\theta_{\text{new}} = \theta + \omega T_s + \frac{1}{2}a_{\text{prev}}T_s^2 + \frac{1}{6}jT_s^3. \quad (44)$$

These formulas are exact if the jerk is constant over the interval.

5.6 Proof of Boundedness

Proposition 5.1 *The trajectory generation algorithm preserves the bounds: $|\omega_{\text{new}}| \leq \max(|\omega|, |\omega_{\text{target}}|)$ (no overshoot beyond target), $|a_{\text{new}}| \leq a_{\text{max}}$, and $|j| \leq j_{\text{max}}$.*

The jerk is explicitly saturated. The acceleration is clamped indirectly: a_{des} is within $[-a_{\text{max}}, a_{\text{max}}]$, and the integration with limited jerk ensures a_{new} stays within $[-a_{\text{max}}, a_{\text{max}}]$ if T_s is small enough (the saturation in the jerk ensures the acceleration does not exceed the limit). The velocity is bounded because if it would cross the target, the algorithm clamps it to the target (common implementation detail). Thus the bounds are satisfied.

6 Relation to Feedback Linearization

Differential flatness is closely related to feedback linearization. In fact, if a system is flat, it can be transformed into a linear system via a change of coordinates and a static state feedback. The flat output and its derivatives serve as a linearizing output.

For the PMSM, the flat output $y = \theta_m$ yields a chain of integrators in the coordinates:

$$z_1 = y = \theta_m, \quad (45)$$

$$z_2 = \dot{y} = \omega_m, \quad (46)$$

$$z_3 = \ddot{y} = \frac{1}{J}(K_t \dot{i}_q - B\omega_m - T_L), \quad (47)$$

$$z_4 = \dddot{y} = \frac{1}{J}(K_t \ddot{i}_q - B\dot{\omega}_m). \quad (48)$$

In these coordinates, the system becomes:

$$\dot{z}_1 = z_2, \quad (49)$$

$$\dot{z}_2 = z_3, \quad (50)$$

$$\dot{z}_3 = z_4, \quad (51)$$

$$\dot{z}_4 = v, \quad (52)$$

where v is a new input related to the original v_q by a nonlinear transformation. This is a linear system (a chain of integrators). Thus flatness implies feedback linearizability.

The advantage of flatness over feedback linearization is that it does not require an explicit state transformation for trajectory planning; we can plan directly in the flat output space.

7 Practical Control Architecture

The theoretical framework leads to the following control architecture:

Step 1: Generate smooth reference trajectory $\omega^*(t), \dot{\omega}^*(t), \ddot{\omega}^*(t)$.

Step 2: Compute feedforward torque T_{ff} using (28).

Step 3: Compute speed PI correction ΔT from speed error.

Step 4: Compute total torque $T^* = T_{ff} + \Delta T$.

Step 5: Compute current reference $i_q^* = T^*/K_t$.

Step 6: Compute current derivative feedforward $\dot{i}_q^*$ using (30).

Step 7: Compute feedforward voltages v_d^*, v_q^* using (31), (32).

Step 8: Measure phase currents and transform to dq : i_d, i_q .

Step 9: Compute current errors $e_d = -i_d$, $e_q = i_q^* - i_q$.

Step 10: Apply current PI corrections: $v_d = v_d^* + \Delta v_d$, $v_q = v_q^* + \Delta v_q$.

Step 11: Apply inverse Park and SVPWM to generate switching signals.

This architecture combines the model-based feedforward (exact for nominal parameters) with feedback for robustness.

8 Conclusion

Differential flatness provides a rigorous mathematical foundation for feedforward control of nonlinear systems. For the PMSM, the rotor mechanical angle is a flat output, and its derivatives up to third order determine all states and inputs. The path generation problem reduces to constructing a smooth velocity trajectory with bounded acceleration and jerk, which is achieved by an online jerk-limited trajectory generator. The resulting control architecture—feedforward plus PI feedback—is both theoretically sound and practically implementable on embedded systems.

The mathematical tools introduced—Lie derivatives, relative degree, and flatness—are essential for understanding the structure of nonlinear control systems and designing high-performance controllers.

A Proof of the Stopping Acceleration Formula

We want to find the maximum acceleration a_{stop} that can be reduced to zero without exceeding jerk limit j_{max} while eliminating a velocity error e (assume $e > 0$). The deceleration phase starts at acceleration a_{stop} and applies jerk $-j_{\text{max}}$ until acceleration reaches zero. The time to reach zero acceleration is $t = a_{\text{stop}}/j_{\text{max}}$. The velocity change during this time is:

$$\Delta\omega = \int_0^t a(\tau)d\tau = \int_0^t (a_{\text{stop}} - j_{\text{max}}\tau)d\tau = a_{\text{stop}}t - \frac{1}{2}j_{\text{max}}t^2 = \frac{a_{\text{stop}}^2}{2j_{\text{max}}}. \quad (53)$$

Setting $\Delta\omega = e$ gives $a_{\text{stop}} = \sqrt{2j_{\text{max}}e}$. This is the maximum initial acceleration from which we can stop within the jerk limit.

B Discrete-Time Integration Derivation

Assume constant jerk j over the sample interval $[0, T_s]$. Then acceleration is $a(t) = a_0 + jt$. Velocity is $\omega(t) = \omega_0 + a_0t + \frac{1}{2}jt^2$. Position is $\theta(t) = \theta_0 + \omega_0t + \frac{1}{2}a_0t^2 + \frac{1}{6}jt^3$. Evaluating at $t = T_s$ yields (42)–(44).